Source Code :-

```
import tensorflow as tf
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

# Load dataset
(x_train, y_train), (x_test, y_test) = tf.keras.datasets.mnist.load_data()

x_train = x_train.reshape(-1, 784).astype(np.float32) / 255.0
x_test = x_test.reshape(-1, 784).astype(np.float32) / 255.0

y_train = tf.one_hot(y_train, 10)
y_test = tf.one_hot(y_test, 10)

# Flexible Training Function
def train_model(hidden_units=128,
                activation="relu",
                learning_rate=0.01,
                batch_size=128,
                epochs=15):

    # Activation selector
    if activation == "relu":
        act_fn = tf.nn.relu
    elif activation == "sigmoid":
        act_fn = tf.nn.sigmoid
    elif activation == "tanh":
        act_fn = tf.nn.tanh
    else:
        raise ValueError("Unsupported activation")

    # Initialize weights
    W1 = tf.Variable(tf.random.normal([784, hidden_units], stddev=0.1))
    b1 = tf.Variable(tf.zeros([hidden_units]))

    W2 = tf.Variable(tf.random.normal([hidden_units, 10], stddev=0.1))
    b2 = tf.Variable(tf.zeros([10]))
```

```python
optimizer = tf.optimizers.SGD(learning_rate)

loss_history = []
acc_history = []

for epoch in range(epochs):

    for i in range(0, x_train.shape[0], batch_size):
        x_batch = x_train[i:i+batch_size]
        y_batch = y_train[i:i+batch_size]

        with tf.GradientTape() as tape:

            z1 = tf.matmul(x_batch, W1) + b1
            a1 = act_fn(z1)

            z2 = tf.matmul(a1, W2) + b2

            loss = tf.reduce_mean(
                tf.nn.softmax_cross_entropy_with_logits(
                    labels=y_batch,
                    logits=z2))

        grads = tape.gradient(loss, [W1, b1, W2, b2])
        optimizer.apply_gradients(zip(grads, [W1, b1, W2, b2]))

    # Accuracy
    z1 = tf.matmul(x_test, W1) + b1
    a1 = act_fn(z1)
    z2 = tf.matmul(a1, W2) + b2
    preds = tf.nn.softmax(z2)

    correct = tf.equal(tf.argmax(preds, 1), tf.argmax(y_test, 1))
    acc = tf.reduce_mean(tf.cast(correct, tf.float32))

    loss_history.append(loss.numpy())
    acc_history.append(acc.numpy())
```

```
        print(f"Epoch {epoch+1}: Loss={loss.numpy():.4f}
TestAcc={acc.numpy():.4f}")

    return acc.numpy(), loss_history, acc_history
```

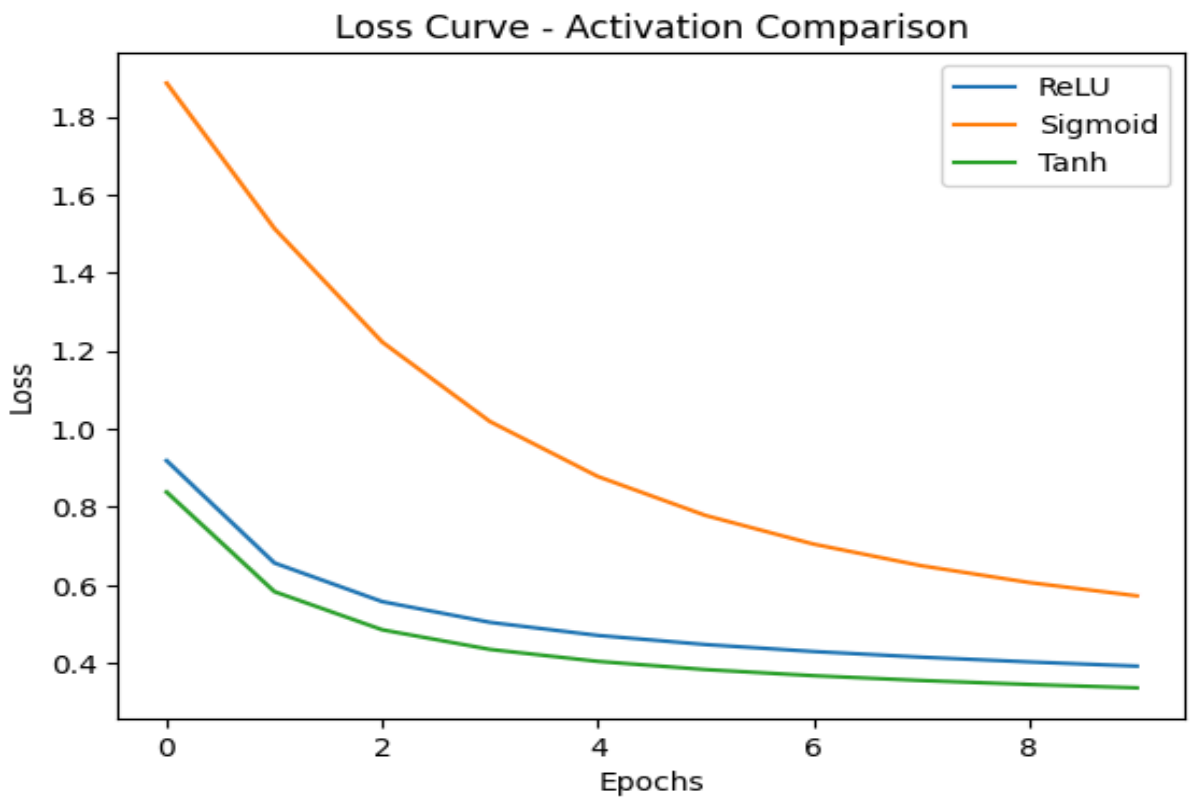
EXPERIMENT 1: Activation Functions

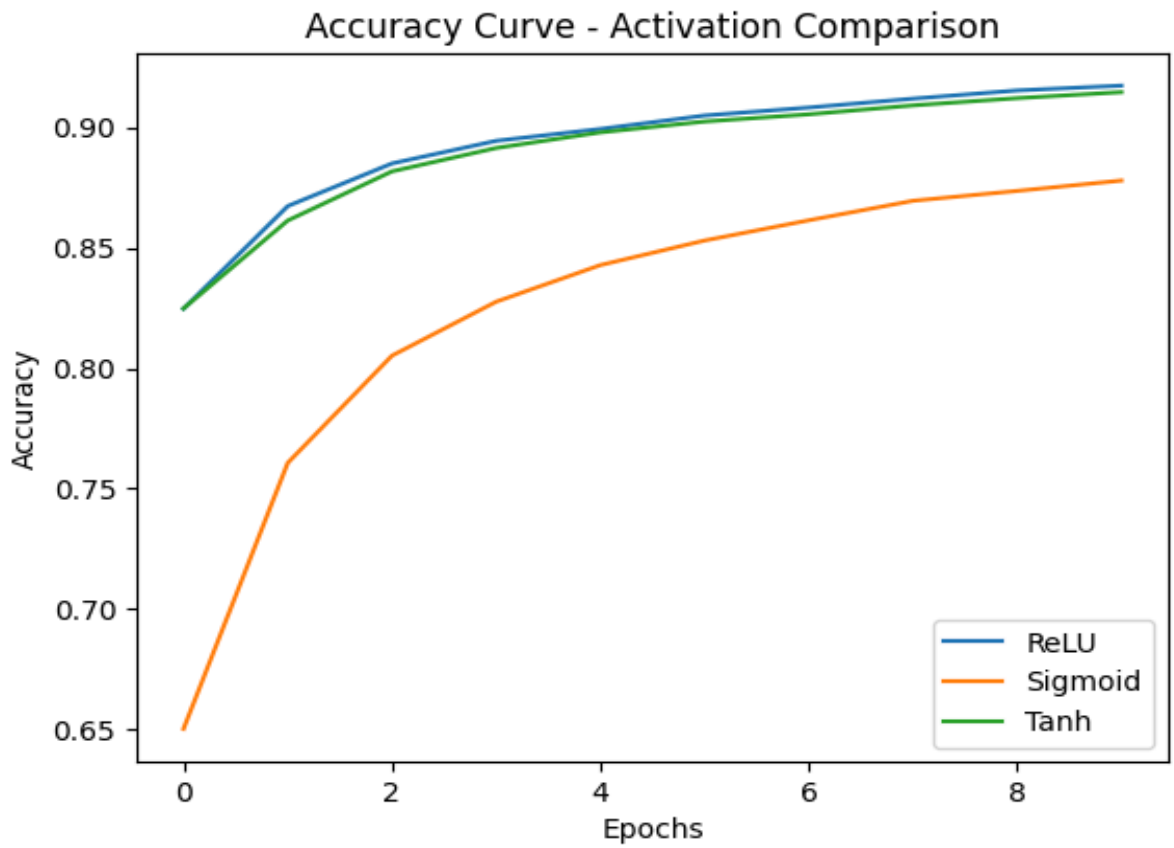
```
# Run experiments
relu_acc, relu_loss, relu_acc_hist = train_model(activation="relu")
sig_acc, sig_loss, sig_acc_hist = train_model(activation="sigmoid")
tanh_acc, tanh_loss, tanh_acc_hist = train_model(activation="tanh")

# -------- Loss Curve --------
plt.figure()
plt.plot(relu_loss, label="ReLU")
plt.plot(sig_loss, label="Sigmoid")
plt.plot(tanh_loss, label="Tanh")
plt.title("Loss Curve - Activation Comparison")
plt.xlabel("Epochs")
plt.ylabel("Loss")
plt.legend()
plt.show()

# -------- Accuracy Curve --------
plt.figure()
plt.plot(relu_acc_hist, label="ReLU")
plt.plot(sig_acc_hist, label="Sigmoid")
plt.plot(tanh_acc_hist, label="Tanh")
plt.title("Accuracy Curve - Activation Comparison")
plt.xlabel("Epochs")
plt.ylabel("Accuracy")
plt.legend()
plt.show()
```

OUTPUT :-


Loss Curve - Activation Comparison
ReLU
Sigmoid
Tanh
Loss
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0
2
4
6
8
Epochs

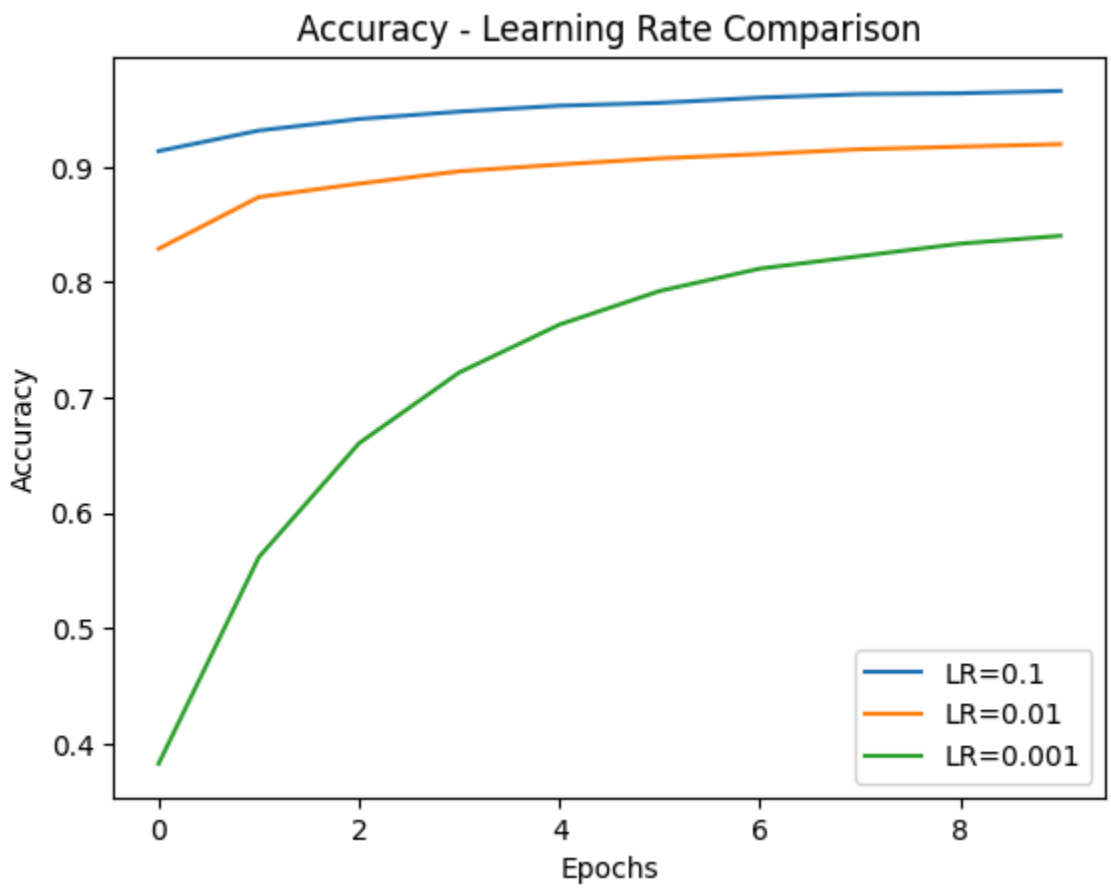

Accuracy Curve - Activation Comparison
ReLU
Sigmoid
Tanh
Accuracy
0.90
0.85
0.80
0.75
0.70
0.65
0
2
4
6
8
Epochs

EXPERIMENT 2: Hidden Layer Size

```
lr1_acc, lr1_loss, lr1_acc_hist = train_model(learning_rate=0.1)
lr2_acc, lr2_loss, lr2_acc_hist = train_model(learning_rate=0.01)
lr3_acc, lr3_loss, lr3_acc_hist = train_model(learning_rate=0.001)

# Accuracy Plot
plt.figure()
plt.plot(lr1_acc_hist, label="LR=0.1")
plt.plot(lr2_acc_hist, label="LR=0.01")
plt.plot(lr3_acc_hist, label="LR=0.001")
plt.title("Accuracy - Learning Rate Comparison")
plt.xlabel("Epochs")
plt.ylabel("Accuracy")
plt.legend()
plt.show()
```

OUTPUT :-

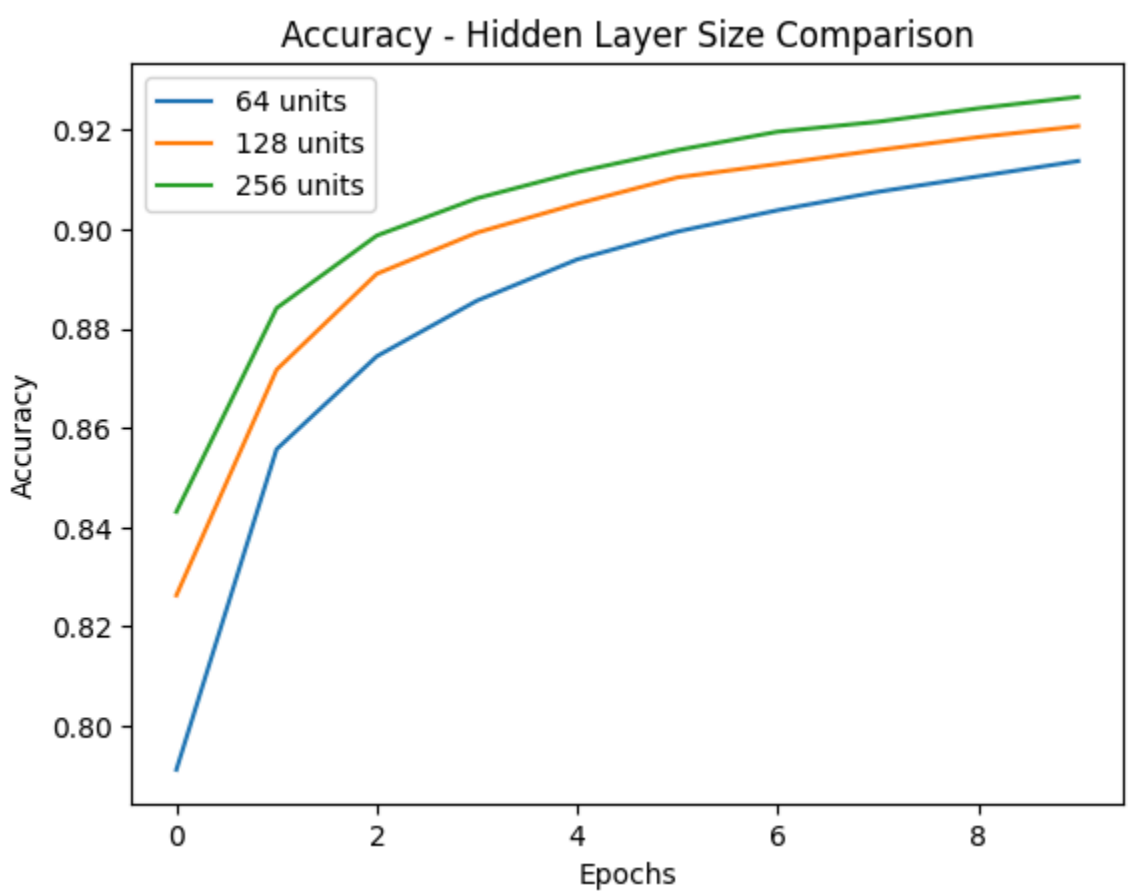


EXPERIMENT 3: Learning Rate

```
h1_acc, h1_loss, h1_acc_hist = train_model(hidden_units=64)
h2_acc, h2_loss, h2_acc_hist = train_model(hidden_units=128)
h3_acc, h3_loss, h3_acc_hist = train_model(hidden_units=256)

plt.figure()
plt.plot(h1_acc_hist, label="64 units")
plt.plot(h2_acc_hist, label="128 units")
plt.plot(h3_acc_hist, label="256 units")
plt.title("Accuracy - Hidden Layer Size Comparison")
plt.xlabel("Epochs")
plt.ylabel("Accuracy")
plt.legend()
plt.show()
```

OUTPUT :-

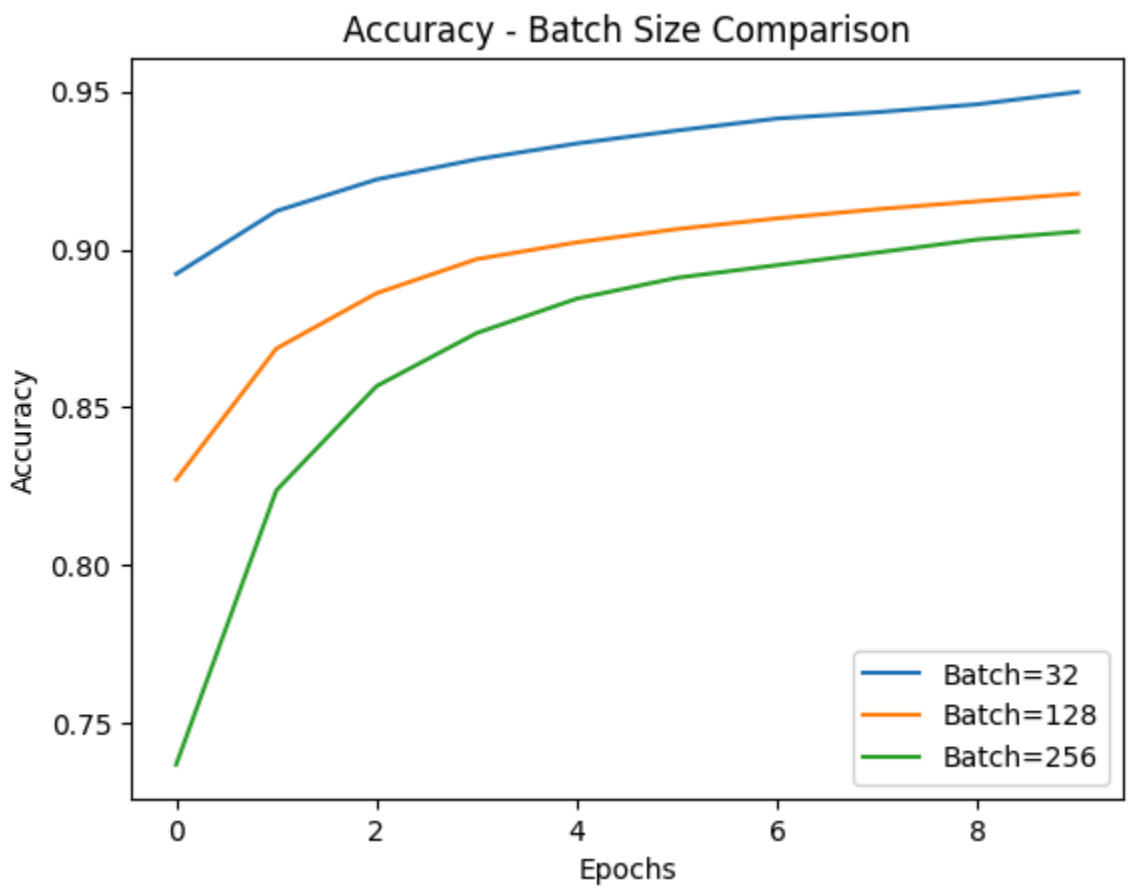


EXPERIMENT 4: Batch Size

```
b1_acc, b1_loss, b1_acc_hist = train_model(batch_size=32)
b2_acc, b2_loss, b2_acc_hist = train_model(batch_size=128)
b3_acc, b3_loss, b3_acc_hist = train_model(batch_size=256)

plt.figure()
plt.plot(b1_acc_hist, label="Batch=32")
plt.plot(b2_acc_hist, label="Batch=128")
plt.plot(b3_acc_hist, label="Batch=256")
plt.title("Accuracy - Batch Size Comparison")
plt.xlabel("Epochs")
plt.ylabel("Accuracy")
plt.legend()
plt.show()
```

OUTPUT :-

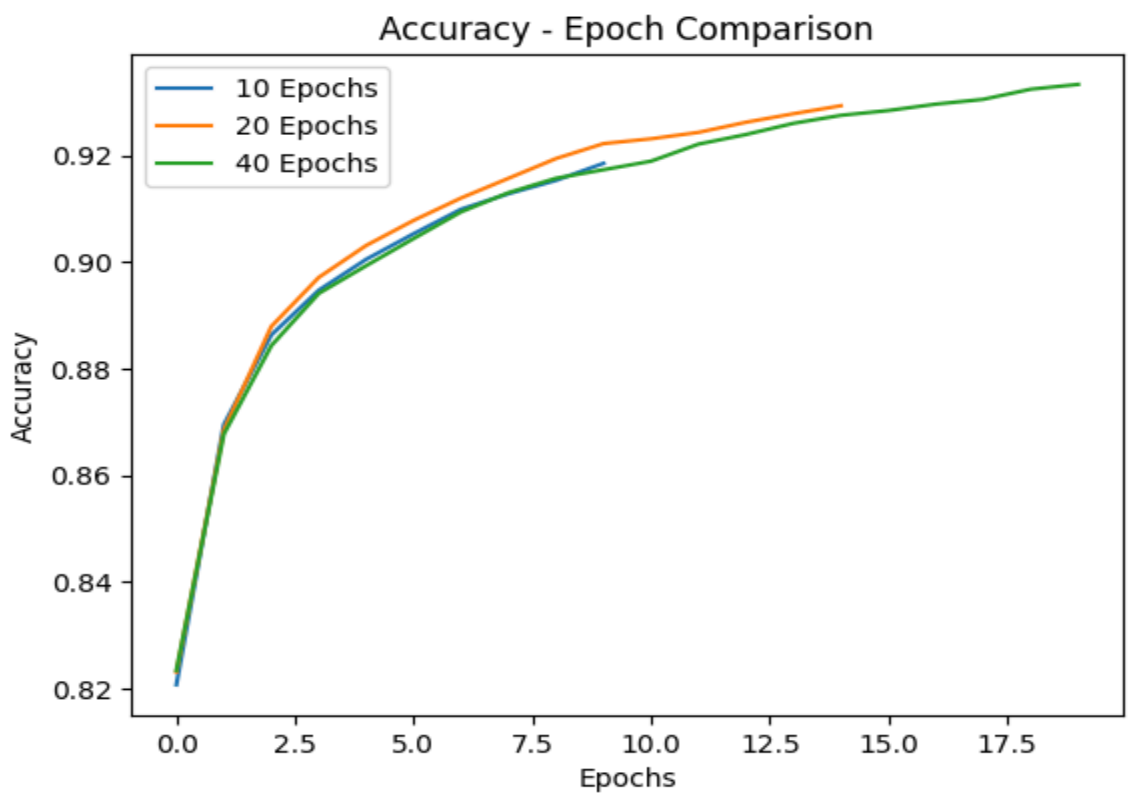


EXPERIMENT 5: Number of Epochs

```
e1_acc, e1_loss, e1_acc_hist = train_model(epochs=10)
e2_acc, e2_loss, e2_acc_hist = train_model(epochs=15)
e3_acc, e3_loss, e3_acc_hist = train_model(epochs=20)

plt.figure()
plt.plot(e1_acc_hist, label="10 Epochs")
plt.plot(e2_acc_hist, label="20 Epochs")
plt.plot(e3_acc_hist, label="40 Epochs")
plt.title("Accuracy - Epoch Comparison")
plt.xlabel("Epochs")
plt.ylabel("Accuracy")
plt.legend()
plt.show()
```

OUTPUT :-



FINAL OUTPUT :-

	Experiment	Final Accuracy
0	ReLU	0.9173
1	Sigmoid	0.8779
2	Tanh	0.9146
3	LR=0.1	0.9657
4	LR=0.01	0.9196
5	LR=0.001	0.8402
6	Hidden=64	0.9137
7	Hidden=128	0.9207
8	Hidden=256	0.9266
9	Batch=32	0.9498
10	Batch=128	0.9176
11	Batch=256	0.9056
12	Epoch=10	0.9185
13	Epoch=20	0.9293
14	Epoch=40	0.9333